

## CN -- Network Topology

[Target] Target: Name all 6 topologies + their single biggest advantage/disadvantage in 30s.  
Read time: 12 minutes

### What Is Network Topology?

A network is formed by physically connecting two or more devices with two or more links.  
Topology = the arrangement/layout of these devices and connections.

Mnemonic (Hinglish): "Bahut Star Ring Tree Mesh Hybrid hai" B S R T M H  
Bus, Star, Ring, Tree, Mesh, Hybrid

### 6 Types of Network Topology

mermaid

graph TD

NT[Network Topology]

NT --> B[Bus]

NT --> S[Star]

NT --> R[Ring]

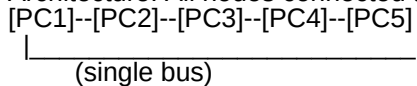
NT --> T[Tree]

NT --> M[Mesh]

NT --> H[Hybrid]

#### 1. Bus Topology

Architecture: All nodes connected to a single backbone cable (the "bus")



#### Feature

Detail

#### Key advantage

Less cabling; one computer failure doesn't affect others

#### Key disadvantage

Backbone cable fails = entire network crashes

#### Congestion

One computer at a time; information goes along cable

#### Reliability

Single point of failure -- common cable fails = all down

#### Security

Any computer can see ALL data on the bus

#### Complexity

Easy to add/remove nodes

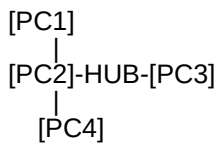
#### Cable

Coaxial, twisted pair, fiber

Memory (Hinglish): "Bus mein ek hi sadak hai -- sadak toot gayi toh sab ruk gaye"  
(Bus has one road -- if the road breaks, everyone stops)

#### 2. Star Topology

Architecture: All peripheral nodes connected to a central node (hub/switch/router)



Feature  
Detail

Key advantage  
Easy error detection; easy to install; one failure doesn't affect others

Key disadvantage  
Hub/switch failure = entire network crashes

Congestion  
Better than bus; signals go only to intended recipient

Reliability  
Hub failure = all down (single point of failure at center)

Security  
Depends on central device security

Complexity  
Average

Cable  
Coaxial, twisted pair, fiber; max 100 meters from hub

Memory (Hinglish): "Star mein sab ke upar ek raja hai -- raja gaya toh sab gaye"  
(In star, everyone has one king -- king falls, all fall)

3. Ring Topology  
Architecture: Each node connected to exactly 2 other nodes, forming a ring. Data travels in one direction (unidirectional).  
[PC1] [PC2] [PC3] [PC4] [back to PC1]

Feature  
Detail

Key advantage  
Equal access for all nodes; adding nodes doesn't degrade performance

Key disadvantage  
One node failure can shut down entire network

Congestion  
Data goes one direction, passes each node until correct one found

Reliability  
Cable or node failure = whole system down

## Security

Data passes from one device to next (each node sees all data)

## Complexity

Simple data flow

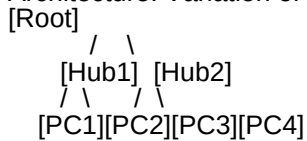
## Cable

Twisted pair; requires more cables than bus

Memory (Hinglish): "Ring mein ek kadi tooti toh poora haar toot gaya"  
(In a ring, one link breaks and the whole necklace breaks)

## 4. Tree Topology

Architecture: Variation of Star -- hierarchical structure, branches like a tree. Root node at top, branches spread out.



## Feature

Detail

## Key advantage

Supported by most hardware/software; scalable

## Key disadvantage

Root node failure = entire network crashes

## Congestion

Transmission from any station propagates to all nodes

## Reliability

Individual node failure doesn't affect hierarchy; root failure = all down

## Security

Data passes over more than one node

## Complexity

Complex (combination of star + bus)

## Cable

Coaxial, Twisted pair, Fiber

## 5. Mesh Topology

Architecture: Every device connected to every other device. For n nodes =  $n(n-1)/2$  communication links.

[PC1]-----[PC2]



[PC4]--[PC3]

For 4 nodes:  $4(4-1)/2 = 6$  links

Feature  
Detail

Key advantage  
Most reliable -- one device failure doesn't break network

Key disadvantage  
Very expensive + very difficult to configure

Congestion  
Minimum -- direct point-to-point links

Reliability  
[OK] Best -- failure of one device doesn't affect others

Security  
Data passes over more than one node

Complexity  
Most complex

Cable  
All types (LAN + WAN)

Memory (Hinglish): "Mesh mein sab se sab jude hain -- ek gira toh koi fark nahi"  
(In mesh, everyone connected to everyone -- one falls, doesn't matter)

6. Hybrid Topology  
Architecture: Combination of two or more different topologies used together.

Feature  
Detail

Key advantage  
Most effective; best features of multiple topologies combined

Key disadvantage  
Most complex to design, install, and configure

Congestion  
Depends on which topologies are combined

Reliability  
Extremely rare complete failure

Security  
Worst (most exposed due to complexity)

Complexity  
The most complicated

Topology Comparison -- Master Table

Feature  
Bus  
Star  
Ring  
Tree  
Mesh  
Hybrid

Structure  
Single cable  
Central hub  
Circle  
Hierarchical  
Full connections  
Mixed

Failure impact  
Whole network  
Whole network  
Whole network  
Only branch (root = all)  
Only that device  
Varies

Single point of failure  
Bus cable  
Hub/Switch  
Any node  
Root node  
None  
Varies

Cost  
Cheap  
Moderate  
Expensive  
Moderate  
Most expensive  
Varies

Reliability  
Low  
Medium  
Low  
Medium  
Highest  
Varies

Complexity  
Easiest  
Easy  
Moderate  
Complex  
Very complex  
Most complex

Security  
Worst  
Moderate

Low  
Low  
Low  
Worst

Links for n nodes

n-1  
n-1  
n  
n-1  
 $n(n-1)/2$   
--

Which Topology to Use? (Quick Guide)

|                           |                        |
|---------------------------|------------------------|
| Small home network        | Star (easy, hub-based) |
| Large enterprise          | Hybrid                 |
| Max reliability needed    | Mesh                   |
| Cost-sensitive            | Bus (but outdated)     |
| Hierarchical organization | Tree                   |

Your 30-Second Script -- Topologies

"There are 6 topologies. Bus uses a single backbone cable -- cheap but whole network fails if bus breaks. Star connects all to a central hub -- easy to manage but hub failure brings everything down. Ring connects each node to two others -- one node failure crashes all. Tree is hierarchical star -- root failure is catastrophic. Mesh connects every device to every other -- most reliable but most expensive. Hybrid combines multiple topologies."

Follow-Up Questions

Q: Which topology is most reliable and why?

Mesh -- because every device has a direct connection to every other device. If one device or link fails, data can still be routed through alternative paths. No single point of failure.

Q: Why is Star topology most commonly used in office networks?

Easy to add/remove devices, easy to troubleshoot (isolate one node without affecting others), and failure of one node doesn't bring down the network. Hub/switch failure is the only risk.

Q: How many links does Mesh topology need for 5 nodes?

$n(n-1)/2 = 5 \cdot 4/2 = 10$  links. This is why mesh is expensive for large networks.

Q: What is the difference between physical and logical topology?

Physical topology = how devices are physically wired/connected. Logical topology = how data actually flows through the network. Example: A ring physical topology can have a bus logical topology.